

Christmas gingerbread (Lebkuchen) and Christmas cheer--review of the potential role of mood elevating amphetamine-like compounds formed in vivo and...

The typical spices used in winter include nutmeg, cinnamon, clove and anise. These spices contain two groups of chemicals, the allylbenzenes and their isomers, the propenylbenzenes. It was suggested 40 years ago by Alexander Shulgin that these substances act as metabolic precursors of amphetamines. The biotransformation of these precursors to nitrogen-containing metabolites is reviewed. These reactions have not been reported in humans. Whether or not the pharmacology and toxicology of spices such as nutmeg can be explained on the basis of their allylbenzene or propenylbenzene content is speculative. Humans may be exposed to amphetamines derived from these precursors in forno, the formation during baking and cooking, for example in the preparation of Lebkuchen, or Christmas gingerbread. It is possible that this may be responsible, in part, for uplifting our mood in winter. However, the role of these aromatic substances, acting simply as odours, evoking old memories of winters past, cannot be ignored. Whether spices have a true pharmacological effect or they act as aromatherapy remains to be elucidated through clinical and laboratory studies.